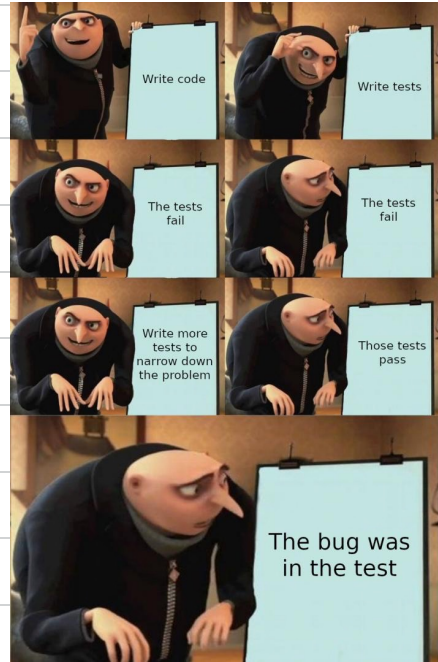




Brenan Keller
@brenankeller

A QA engineer walks into a bar.
Orders a beer. Orders 0 beers.
Orders 999999999999 beers.
Orders a lizard. Orders -1 beers.
Orders a ueicbksjdhd.

First real customer walks in and asks where the bathroom is. The bar bursts into flames, killing everyone.



Agenda

① TESTS!! Wait what??
What's that?!

② What is CI/CD



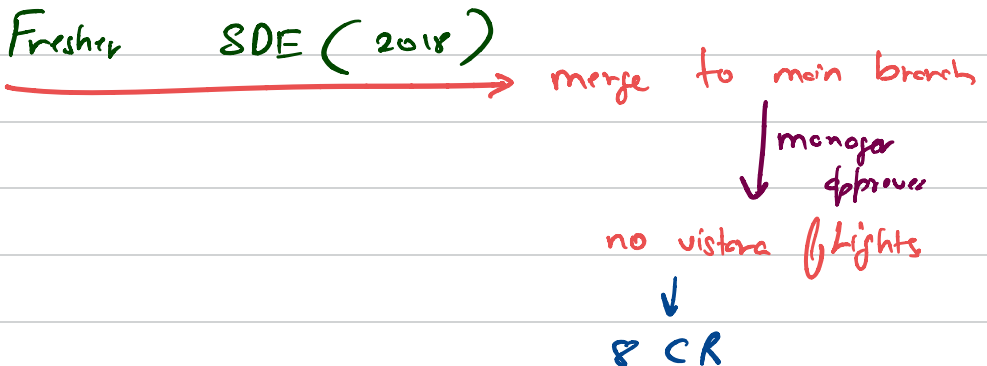
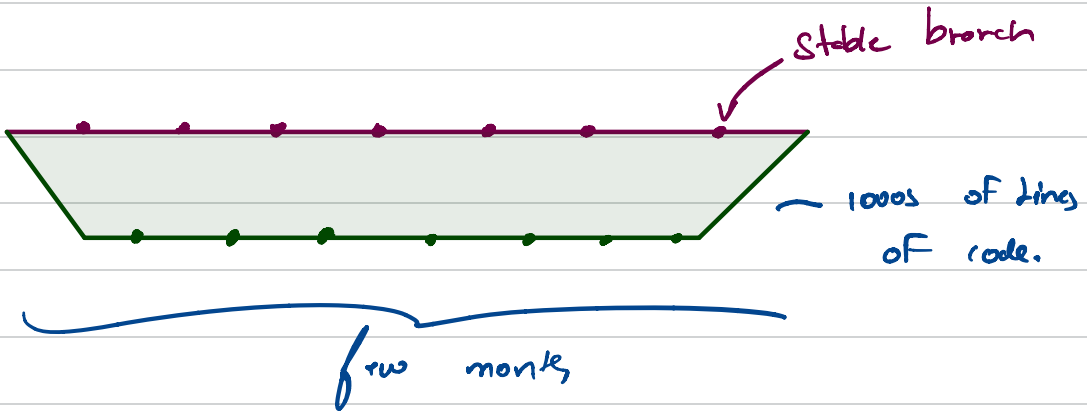
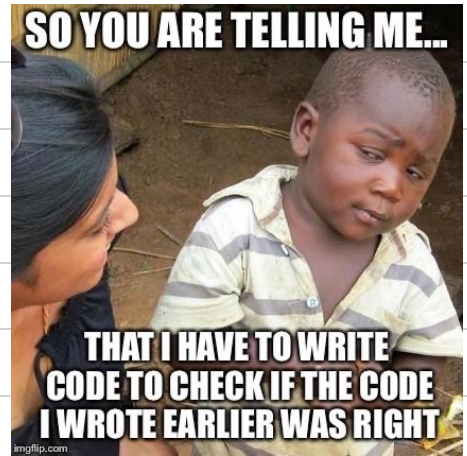
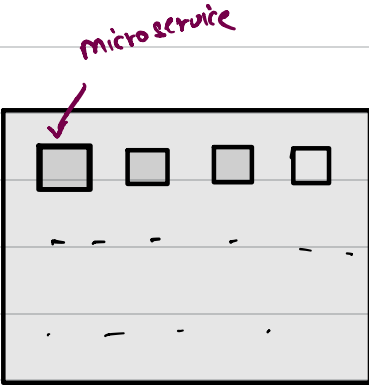
Dave Dustin
@venzann

Genie: I'll give you one billion dollars if you can spend 100M in a month. There are 3 rules: No gifting, no gambling, no throwing it away.

SRE: Can I use AWS?

Genie: There are 4 rules.

TESTS

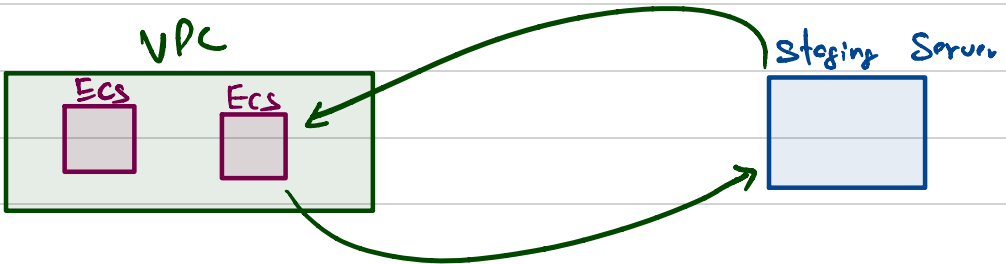


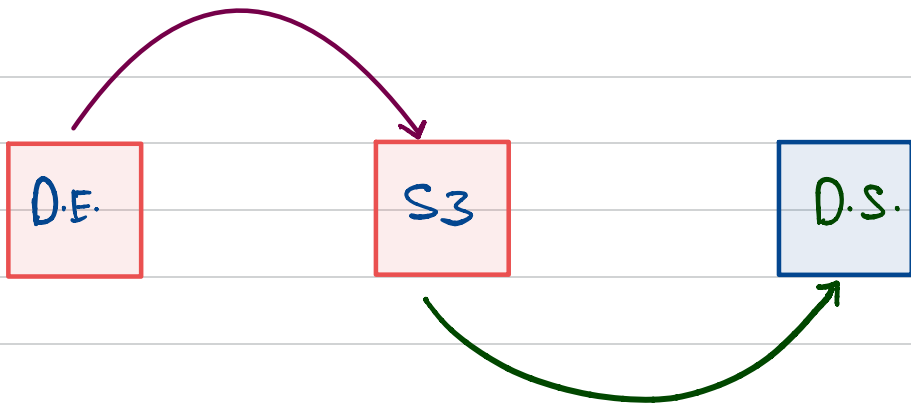


What are some core objectives of testing??

1. Entire codebase should be still functional.
2. Any online/offline metric should not be hampered.
3. Latency should not increase/decrease in an unexpected way.
4. Data Integrity should not change or be compromised.

Latency = Time taken for request to arrive at
the remote server. — 90ms
+
Time taken for TTL model to respond, — 20ms
+
Time taken for response to arrive at
the user's computer — 90ms



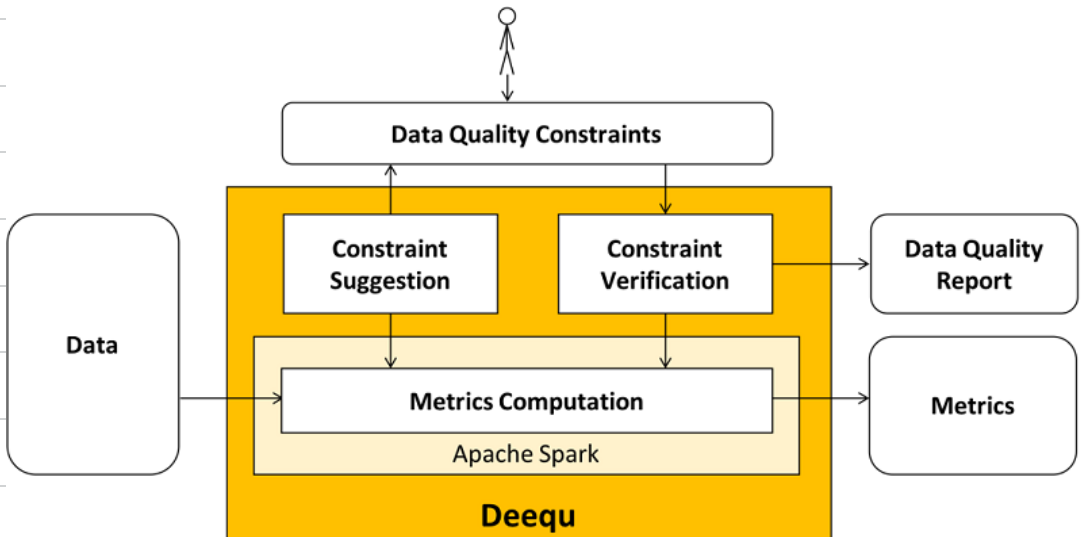


Dynamic Discounting — stale data

CTR $\rightarrow$ $\downarrow$ %
Profit no. $\downarrow$

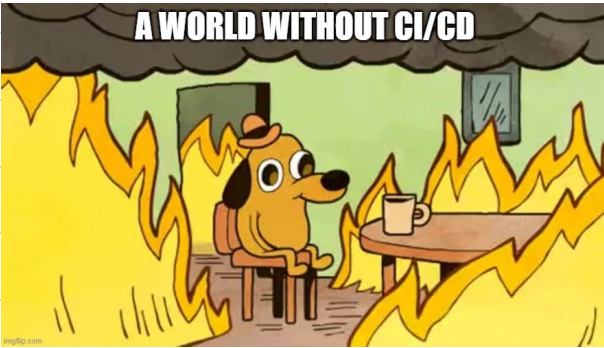
$f_i \rightarrow 3\% \rightarrow 20\%$
% of null values

<https://aws.amazon.com/blogs/big-data/test-data-quality-at-scale-with-deequ/>



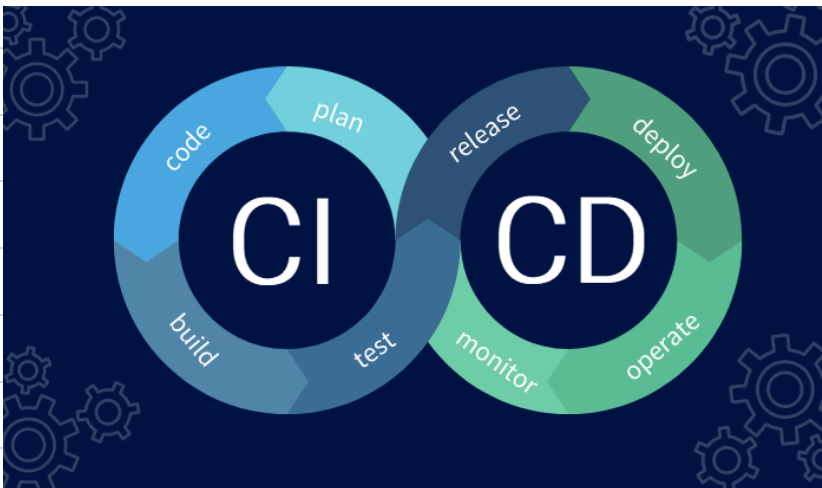
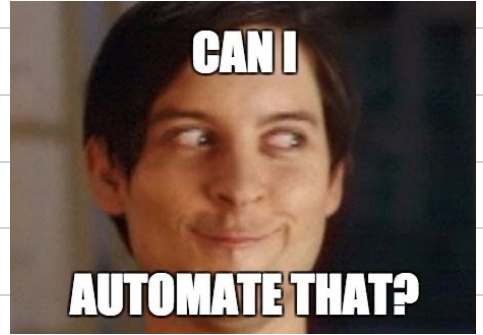
CI/CD

A WORLD WITHOUT CI/CD



CAN I

AUTOMATE THAT?



How does Continuous Delivery (CD) differ from Continuous Integration (CI)?

0 users have participated

A

CD focuses on user interface improvements

0%

B

CD automates the delivery of completed code to testing and development environments

0%

C

CI automates the deployment of code changes to production

0%

D

CI and CD are interchangeable terms

0%

End Quiz Now

V

2

Veeresh

1/1

88.70

J

1

J. Ramesh Reddy

1/1

89.60

S

3

SAGAR GUPTA

1/1

87.44

4

P

Prashant Barkale

1/1

85.77

5

J

Jigar

1/1

79.83

6

R

Rishikesh Kumar Jha

1/1

78.07

7

A

Anirban

1/1

71.00

8

M

Manikandan A

1/1

69.30

9

D

Deepa Krishna

1/1

64.50

What is the primary objective of Continuous Integration (CI) in software development?

0 users have participated

A

Deploying code changes directly into production

0%

B

Automating code delivery to testing environments

0%

C

Identifying and testing code changes frequently

0%

D

Scheduling periodic software updates

0%

End Quiz Now

P

2

Prashant Barkale

2/2

166.43

J

1

J. Ramesh Reddy

2/2

180.80

J

3

Jigar

2/2

160.57

4

M

Manikandan A

2/2

158.53

5

V

Veeresh

1/2

88.70

6

S

SAGAR GUPTA

1/2

87.44

7

R

Rishikesh Kumar Jha

1/2

78.07

8

A

Anirban

1/2

71.00

9

D

Deepa Krishna

1/2

64.50

What is the primary objective of a unit test?

0 users have participated

A

Testing an entire application as a single unit

0%

B

Verifying the behavior of an application module in isolation

0%

C

Analyzing the overall performance of an application

0%

D

Ensuring compatibility with external systems

0%

End Quiz Now

M

2

Manikandan A

3/3

251.83

J

1

J. Ramesh Reddy

3/3

267.06

P

3

Prashant Barkale

3/3

251.60

4

J

Jigar

3/3

241.83

5

Y

Yadwinder kaur

1/3

90.40

6

V

Veeresh

1/3

88.70

7

S

SAGAR GUPTA

1/3

87.44

8

R

Rishikesh Kumar Jha

1/3

78.07

9

A

Anirban

1/3

71.00

10

D

Deepa Krishna

1/3

64.50

What are the three key phases typically found in a unit test?

0 users have participated

A

Initializing, executing, and debugging

0%

B

Setup, test execution, and result verification

0%

C

Coding, compiling, and deploying

0%

D

Documentation, validation, and optimization

0%

End Quiz Now

P

2

Prashant Barkale

4/4

331.73

J

1

J. Ramesh Reddy

4/4

356.16

J

3

Jigar

4/4

325.10

4

M

Manikandan A

4/4

321.20

5

S

SAGAR GUPTA

2/4

179.62

6

A

Anirban

2/4

141.73

7

Y

Yadwinder kaur

1/4

90.40

8

V

Veeresh

1/4

88.70

9

R

Rishikesh Kumar Jha

1/4

78.07

10

D

Deepa Krishna

1/4

64.50

Steps for automating tests:

1. Decide OS: Ubuntu, Mac, or Windows?
2. Install python, and requirements.
3. Perform tests!
4. [Push to Dockerhub](#)